

Roll No: 101,102,105,106,107

Prof: Mam Hafsa Nazir

Subject: Research In Education

Assignment:

Topic:

Inferential of Data Analysis

Inferential Statistics:

Inferential statistics helps study a sample of data and make conclusions about its population. The purpose of studying Inferential statistics is to infer the behavior of a population.

When you have collected data from a **sample**, You can use inferential statistics to understand the larger population from which the sample is taken.

- The procedure includes choosing a sample, applying tools like regression analysis testes, and making judgments using logical reasoning.

Example:

You randomly select a sample of 11th graders in your state and collect data on their scores and other characteristics. You can use

Whole population of 11th graders in the state based on your sample data

Types of Inferential Statistics:

1. Hypothesis Testing
2. Confidence Interval
3. Regression Analysis

Hypothesis Testing

Meaning:

Hypothesis is a testable statement. It is formulated after literature review. It explains the behaviors or events. The researcher does not try to prove the hypothesis rather he collects data that either support it. All the hypothesis are based on previous knowledge. They are aimed at extending knowledge. Every aspect of research e.g. samples, instruments, design, procedure, data analysis techniques and conclusions are affected by hypothesis.

Definitions:

“ A hypothesis is a tentative explanation for certain behaviors, phenomena or the events that have occurred or will occur.
(L.R. Gay)

Definition of Hypothesis Testing:

Hypothesis testing is a formal procedure for investigating our ideas about the world using

statistics. It is most often used by scientists to test specific predictions, called hypothesis testing, that arise from theories.

Steps:

Testing a hypothesis in educational research typically involves several steps:

1) Formulate the Hypothesis:

Clearly state your research question and the hypothesis you want to test. **For example**, if you're investigating the effect of a teaching method on students performance, your hypothesis might be: "Students who receive Method A instruction will perform better than those who receive Method B instruction." After developing your initial research hypothesis (the prediction that you want to investigate), it is important to restate it as a null (H_0) and alternate (H_a) hypothesis so that you can test it mathematically.

The alternate hypothesis is usually your initial hypothesis that predicts a relationship between variables. The null hypothesis is a prediction of no relationship between the variables you are interested in.

2) Select a Research Design:

Choose the appropriate research design, such as experimental, quasi-experimental, or observational, based on your research question and available resources.

3) Select a Sample:

Determine the population you want to study and select a representative sample from that population. Random sampling or stratified sampling methods are often used to ensure the sample is unbiased.

4) Collect Data:

Collect data from your sample using appropriate research instruments, such as surveys, tests, or observations. Ensure the data collection process is standardized and reliable.

5) Analyze Data:

Use statistical methods to analyze the collected data. Depending on your research design and data type, you may use regression analysis, or other statistical tests to compare conclusions.

6) Set a Significance Level:

Decide on the level of significance (e.g., 0.05) to determine whether the results are statistically significant. This is often referred to as alpha(α).

7) Perform the Test:

Conduct the statistical test based on your chosen significance level. This will help you determine if there is enough evidence to either accept or reject your null hypothesis.

8) Interpret Results:

Examine the test results and interpret them in the context of your hypothesis. If the p- value is less than your chosen significance level (α), you can reject the null hypothesis in favor of the alternative hypothesis.

9) Draw Conclusions:

Based on your analysis, draw conclusions about whether your hypothesis is supported or not. Explain the practical implications of your findings.

10) Report Findings:

Communicate your research findings through a research report or academic paper. Include details about your methodology, results, and any limitations of the study.

Confidence Interval

Meaning:

The confidence level represents the proportion (frequency) of acceptable confidence intervals that contain the true value of the unknown parameter. In other terms, the confidence intervals are evaluated using the given confidence level from an endless number of independent samples. So that the proportion of the range contains the true value of the parameter that will be equal to the confidence level.

Mostly, the confidence level is selected before examining the data. The commonly used confidence level is 95% confidence level. However,

other confidence levels are also used, such as 90% and 99% confidence levels.

Definition:

An interval of values computed from the sample, that is almost sure to cover the true population value.

Confidence Interval Formula:

The confidence interval based on the mean and **standard deviation**. Thus, the formula to find CI is

$$\bar{X} \pm Z_{\alpha/2} \times [\sigma / \sqrt{n}]$$

Where

$\bar{X}$ = Mean

Z = Confidence coefficient

α = Confidence level

σ = Standard deviation

N = Sample space

The value after the $\pm$ symbol is known as the margin of error.

Confidence Interval Table:

The confidence interval table for Z values are given as follows

Confidence Interval	Z Value
80%	1.282
85%	1.440
90%	1.645
95%	1.960
99%	2.576
99.5%	2.807
99.9%	3.291

How to Calculate Confidence Interval ?

To calculate the confidence interval, go through the following procedure.

Step 1:

Find the number of observations n (sample space), mean $\bar{X}$, and the standard deviation σ .

Step 2:

Decide the confidence interval of your choice. It should be either 95% or 99%. Then find the Z value for the corresponding confidence interval given in the table.

Step 3:

Finally, substitute all the values in the formula.

Confidence Interval Example:

Question: In a tree, there are hundreds of apples. You are randomly choosing 46 apples with a mean of 86 and a standard deviation of 6.2. Determine that the apples are big enough.

Solution:

Given: Mean, $\bar{X} = 86$

Standard deviation, $\sigma = 6.2$

Number of observations, $n = 46$

Take the confidence level as 95%

Therefore, the value of $z = 1.960$ (from the table)

The formula to find the confidence interval is

$$\bar{X} \pm Z_{\alpha / 2} \times [\sigma / \sqrt{n}]$$

Now , substitute the values in the formula, we get

$$86 \pm 1.960 \times [6.2 / \sqrt{46}]$$

$$86 \pm 1.960 \times [6.2 / 6.78]$$

$$86 \pm 1.960 \times 0.914$$

$$86 \pm 1.79$$

Here, the margin of error is 1.79

Therefore, all the hundreds of apples are likely to be between in the range of 84.21 and 87.79.

Regression Analysis

Meaning:

Regression is the study of the nature of relationship between the variables so that one may be able to predict the unknown value of another variable,

In regression, one variable is considered as an independent variable and another variable is taken as dependent variable.

With the help of regression, possible values of the dependent variable are estimated on the basis of the values of the independent variable.

Definition:

Regression is the measure of the average relationship between two or more variables. (M.M. Blair)

Utility Of Regression

1. Nature of Relationship:

Regression analysis explains the nature of relationship between two variables.

2. Estimation of Relationship:

The mutual relationship between two or more variables can be measured easily by regression analysis.

3. Prediction:

By regression analysis, the value of a dependent variable can be predicted on the basis of the value of an independent variable. **For example,** if price of a commodity rises, what will be the probable fall in demand, this can be predicted by regression.

4. Useful in Economic and Business Research:

Regression analysis is very useful in business and economic research. With the help of regression, business and economic policies can be formulated.

Types Of Regression Analysis:

There are following types of the regression analysis:

1. Simple Regression:

In simple regression analysis, we study only two variables at a time, in which one variable is dependent and other is independent. The relationship between income and expenditure is an example of simple regression.

2. Multiple Regression:

On the contrary, we study more than two variables at a time in multiple regression analysis (at least three variables) in which one is dependent variable and others are independent variable. The study of effect of rain and irrigation on yield of wheat is an example of multiple regression.

3. Linear Regression:

When one variable changes with other variable in some fixed ratio, this is called as linear regression. Such type of relationship is depicted on a graph by means of a straight line.

4. Non-Linear Regression:

On the contrary, when one variable varies with other variable in a changing ratio, then it is referred to as curve- linear non- linear regression. This relationship, expressed on a graph paper takes the form of a curve.

5. Partial Regression:

When two or more variables are studied for functional relationship but at a time, relationship between only two variables is studied and other variables are held constant, then it is known as partial regression.

6. Total Regression:

In total regression all variables are studied simultaneously for the relationship among them.

Report Writing

Meaning:

Report writing is a formal style of writing elaborately on a topic. The tone of a report and report writing format is always formal. The important section to focus on is the target audience. **For example-** report writing about a school event, report writing about a business case, etc. Report writing is written in past tense and according to a definite format.

Format:

Format refers to the general pattern of organization and arrangement of the report.

Style:

Style refers to the rules of spelling, capitalization, punctuation, and typing followed in preparing the report.

Definition:

Research report is a brief description of the research work done by the researcher. It involves several steps to present the report in the form of thesis or dissertation.

Characteristics:

There are following characteristics of the writing format:

- Report should be free from grammatical mistakes.
- Report must present the logical analysis of the subject matter.
- It must be an attempt to solve some intellectual problems.
- Index must be prepared & appended at the end.
- Report must be attractive in appearance, neat & clean whether typed or printed.
- It should state the policy implications relating to the problem under consideration.
- Appendices should be enlisted in respect of all the technical data in the report.
- Bibliography of sources consulted is a must for a good report.
- A research report should be written in past tense.

Format Of a Report:

- Although, there is no set report writing format, however, there are general sections that should be included.
- Unlike essays, reports are written in sections with headings and sub-headings, which are usually numbered.
- There are numerous possible formats available for writing a report, and it mostly depends on the context of topic.

Below given are the components of a report in which they would occur:

1. Title page:

It should include the title, your name and the name of the tutor to whom it is being submitted, date of submission, your course / department. The logo of the organization should also be printed.

2. Acknowledgements:

A list of people and organizations who have helped you in the compilation of report and other related work.

3. Contents page:

A clear, well- formatted list of all the sections and sub- sections of the report. Page numbers should be marked correctly.

4. Abstract:

A summary of the major points, conclusions, and recommendations should be written to give a general overview of report.

5. Introduction:

The first page of the report needs to have an introduction. You will explain the problem and show the reader why the report is being made.

6. Body:

This is the main section of the report. There needs to be several section, with each having a subtitle. The various

sections include Review of Literature, Materials and Methods and Results. A discussion section also included at the end of the body to go over by findings and their significance.

7. Conclusion:

A conclusion should draw out the implications of your findings, with deduction based on the facts described in your main body. The significance and relevance of study discussed in this section.

8. References:

This is list giving the full details of all the sources to which you have made reference within your text.

Relevance Of Report Writing:

1. Report as a Means Of Internal Communication:

- A report act as an effective means of communication within the organization.
- It provides feedback to employees.
- It is prepared for the information and guidance of others connected with the matter / problem.

2. Report Facilitates Decision Making And Planning:

- Report provide reliable data which can be used in the planning and decision making process.
- It acts as a treasure house of reliable information for long term planning and decision making.

3. Report Discloses Unknown Information:

- Report provide information, which may not be known previously.
- The committee members collect data, draw conclusions and provide information which will be new to all concerned parties.
- Even new business opportunities are visible through unknown information available in the reports.

4. Report Gives Information To Employees:

- Report are available to managers and department for internal use.
- They are widely used by the departments for guidance.
- Report provide a feedback to employees and are useful for their self- improvement.

Referencing

Meaning:

In educational research, a reference typically refers to a bibliographic entry that acknowledges the source of information or ideas used in a research paper, thesis, or publication.

Reference Section:

This section includes bibliography, appendix and index.

i. Bibliography / Reference:

This section lists all the sources. The sources are written alphabetically by authors last name that were directly used in writing the report. Every source cited in the research report must be included in the references and every entry listed in the references must appear in the paper. If APA manual is followed, secondary sources are not included in the references.

ii. Appendix:

This section include information and data related to the study which either are not important enough or are too lengthy. Appendix contain materials like tests, questionnaires and cover letters, lists and raw data.

iii. Index:

It enlists all the items of importance alphabetically and the page on which they found.

The last section of the research report deals with bibliography and reference. The important points that must be considered while writing the references are as under:

1. List all the sources alphabetically by author's last name.
2. Every source cited in the paper must be included in the references.
3. Every entry listed in the references must appear in the paper.
4. Primary sources are preferred.

Bibliography Style for a book:

1. Surname of author, give name or two initials.
2. Year of publication.
3. Title of the book (underline or italicized).
4. Edition of more than one.
5. Volume of more than one.
6. Place of publication.
7. Publisher.